B. K. Birla College of Arts, Science and Commerce, Kalyan (W)

TYBSc Physics, Semester VI Special Theory of Relativity

Assignment 2

Unit 1: Lorentz Transformation, Velocity Addition, Aberration, Doppler Effect and Relativistic Dynamics

Name: _____ Roll No.: _____ Date: _____

Total marks: 60 **Each question carries 5 marks**

Instructions: Attempt all questions. Section A: theory (write to the point, with neat diagrams where useful). Section B: numerical problems and derivations (state formulae used, show all steps and give units). Sign convention for the interval: $s^2 = \Delta x^2 + \Delta y^2 + \Delta z^2 - c^2\Delta t^2$; four-vectors $a^\mu = (a^0, \mathbf{a})$ with $a \cdot b = -a^0 b^0 + \mathbf{a} \cdot \mathbf{b}$.

Section A: Theory (6 × 5 = 30 marks)

Q1. Why must the Galilean transformation be replaced in special relativity? Write the Lorentz transformation equations and their inverse, and discuss their main features: linearity, the Galilean limit, the mixing of space and time, and the speed of light as a limiting speed. **[5]**

Q2. Explain the relativistic law of addition of velocities. Show qualitatively why "adding" two speeds less than c can never give a speed greater than c, and why a light signal has speed c in every frame. How does this law explain the result of Fizeau's experiment? **[5]**

Q3. What is aberration of light? Explain Bradley's stellar aberration and the relativistic "headlight" (beaming) effect for a fast-moving source. **[5]**

Q4. Explain the relativistic Doppler effect for light. Distinguish between the longitudinal and the transverse Doppler effects, and explain why the transverse effect has no classical analogue. Give two applications. **[5]**

Q5. Explain why the Newtonian definition of momentum must be modified in relativity. Define the four-velocity and four-momentum of a particle and explain why conservation of four-momentum, if true in one inertial frame, is true in all. **[5]**

Q6. Explain the principle of mass–energy equivalence with suitable examples (nuclear binding energy, fusion in the Sun, pair annihilation). Why was the change of mass in chemical reactions never noticed? **[5]**

Section B: Numericals and Derivations (6 × 5 = 30 marks)

Q7. Derive the Lorentz transformation equations for two frames in the standard configuration and obtain the inverse transformation. **[5]**

Q8. An event occurs in frame S at $x = 4.0\,\text{km}$, $t = 20\,\mu\text{s}$. Frame S' moves at $0.6c$ along x. Find x' and t', and verify that $x^2 - c^2t^2 = x'^2 - c^2t'^2$. **[5]**

Q9. Derive the relativistic velocity-addition formula for velocities along the x-axis. Two particles move in opposite directions, each with speed $0.75c$ relative to the laboratory. Find the speed of one relative to the other and compare with the classical answer. **[5]**

Q10. Derive the formula for the longitudinal relativistic Doppler effect. The Hβ line (486.1 nm) in the spectrum of a galaxy is observed at 510.4 nm. Find the speed of the galaxy and state whether it is approaching or receding. **[5]**

Q11. Show that $E^2 = p^2c^2 + m^2c^4$ for a free particle. An electron is accelerated from rest through a potential difference of 2.0 MV. Find its total energy, γ, speed and momentum (in MeV/c). **[5]**

Q12. A proton moves along the x-axis of frame S with total energy 2.0 GeV. (a) Find its momentum in GeV/c. (b) Using the transformation of momentum and energy, find its energy and momentum in a frame moving at $0.5c$ along x. (c) Verify that $E^2 - p^2c^2$ is the same in both frames. **[5]**

Useful constants: $c = 3.0 \times 10^8$ m/s; $e = 1.6 \times 10^{-19}$ C; $m_e c^2 = 0.511$ MeV; $m_p c^2 = 938.3$ MeV; $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m; $\mu_0 = 4\pi \times 10^{-7}$ H/m; $1/(4\pi\varepsilon_0) = 9.0 \times 10^9$ N m^2/C^2; $G = 6.67 \times 10^{-11}$ N m^2/kg^2; $M_\odot = 1.99 \times 10^{30}$ kg; 1 ly $= 9.46 \times 10^{15}$ m; 1 pc $= 3.086 \times 10^{16}$ m $= 3.26$ ly; 1 yr $= 3.156 \times 10^7$ s; $b_{\rm Wien} = 2.898 \times 10^{-3}$ m K; $k_B = 8.62 \times 10^{-5}$ eV/K; $a = 7.566 \times 10^{-16}$ J m^{-3} K^{-4}.